

OBYEDUL HAQUE EFTY

Electrical & Electronic Engineering Graduate

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RESEARCH & ACADEMIC PROFILE

Electrical and Electronic Engineering graduate with an outstanding academic record (CGPA 3.97/4.00, Summa Cum Laude with Gold Medal) and currently pursuing an M.Sc. in Electrical and Electronic Engineering at AIUB. Passionate about research-driven teaching, student mentorship, and academic development. Experienced in mentoring 50+ EEE freshmen and supporting practical, project-based learning. Active researcher with two peer-reviewed IEEE Texas Power and Energy Conference (TPEC) publications in 2026, indexed in IEEE Xplore, with additional research manuscripts in progress. Strong expertise in electrical power systems, smart grid, renewable energy, VLSI, embedded systems, and digital design, complemented by proficiency in MATLAB/Simulink, Cadence Virtuoso, Xilinx Vivado, CST Microwave Studio, COMSOL Multiphysics and related engineering tools.

EDUCATION

American International University-Bangladesh (AIUB)

Master of Science in Electrical & Electronic Engineering (M.Sc. in EEE)

Dhaka, Bangladesh

Feb 2026 — Present

- **Status:** Ongoing (3rd Semester) | Supervisor: **Dr. Md. Rifat Hazari** (Associate Professor, EEE, AIUB)
- **Focus Areas:** Renewable Energy Technology, Power Systems & Applied Artificial Intelligence (AI)

American International University-Bangladesh (AIUB)

Bachelor of Science in Electrical & Electronic Engineering (B.Sc. in EEE)

Dhaka, Bangladesh

Jan 2022 — Jan 2026

- **Academic Distinction:** CGPA: 3.97 / 4.00 — Summa Cum Laude (Gold Medalist)
- **Curricular Record:** Completed 148 credits; awarded AIUB Dean's List Honors in all 4 academic years (2022–2025) with a continuous 70% merit scholarship.

Mymensingh Govt. College

Higher Secondary Certificate (HSC) — Science Group

Mymensingh, Bangladesh

Aug 2018 — Dec 2020

- **Academic Result:** GPA: 5.00 / 5.00 (Board of Intermediate & Secondary Education, Mymensingh)

Premier Ideal High School

Secondary School Certificate (SSC) — Science Group

Mymensingh, Bangladesh

Jan 2016 — May 2018

- **Academic Result:** GPA: 5.00 / 5.00 (Board of Intermediate & Secondary Education, Dhaka)

TEACHING INTERESTS

Foundational & Core Theory Courses:

- **Circuits & Electronics:** Electrical Circuits I (DC) & II (AC), Electronic Devices & Analog Circuits.
- **Power & Energy Systems:** Renewable Energy Technology, Power System Analysis, High Voltage Transmission Systems, Control Systems, Energy Conversion I & II.
- **Digital Systems & Computing:** VLSI, Digital Logic Design, Microprocessor and Embedded Systems.

Laboratory & Sessional Instruction (Hands-on / Simulation):

- **Hardware & Circuit Labs:** Electric Circuits Lab, Electronics Lab, Electrical Machines Lab.
- **Software & Simulation Tools:** MATLAB/Simulink (Control, Power & Signal Processing), NI Multisim, Proteus, Cadence Virtuoso (IC Layout), and Xilinx Vivado (Verilog/FPGA).

Pedagogical Approach: Committed to Outcome-Based Education (OBE) frameworks, interactive problem-solving tutorials, continuous student advising, and bridging theoretical concepts with hardware-in-the-loop and simulation-based engineering projects.

PUBLICATIONS & PEER-REVIEWED RESEARCH

Peer-reviewed papers published in the conference proceedings of the IEEE Texas Power and Energy Conference (TPEC), Texas A&M University, USA (2026):

1. M. S. Hossain, A. A. Asif, T. Ahmed, A. Al Mujahid, **O. H. Efty**, M. S. Paran, T. Islam, N. Hannan, and A. Shufian, "Adaptive Grid-Forming Control with Hybrid Fault Ride-Through for Transient Stability in Inverter-Dominant Microgrids," in *Proceedings of the 2026 IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, 2026. DOI: [10.1109/TPEC67884.2026.11513209](https://doi.org/10.1109/TPEC67884.2026.11513209)

2. M. K. Rahat, A. Hossain, T. Ahmed, A. A. Asif, **O. H. Efty**, A. Al Mujahid, Z. Hasan, A. Rahman, N. Hannan, and A. Shufian, “Enhancing the Reliability of Gear-Train Fault Diagnosis Using Power Spectral Density Analysis of Motor Current Signals,” in *Proceedings of the 2026 IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, 2026. DOI: [10.1109/TPEC67884.2026.11513044](https://doi.org/10.1109/TPEC67884.2026.11513044)

Research Interests: Smart Grid Technologies | Microgrids and Microgrid Stability | Renewable Energy Technology Integration | Applied AI in Power Systems | Photonics | Semiconductor Device Physics & Nanoelectronics

ACADEMIC MENTORING & EXTRACURRICULAR ACTIVITIES

AIUB Electrical & Electronic Engineering Department
Peer Academic Mentor

Dhaka, Bangladesh
Oct 2023 — Dec 2025

- Instructed and mentored a cohort of 50+ EEE freshmen in foundational electrical engineering concepts, including electrical circuit analysis, introductory digital logic design, and basic programming modules.
- Formulated custom problem sets and guided weekly interactive laboratory and tutoring sessions to bridge theoretical classroom principles with practical, hardware-in-the-loop engineering projects.
- Achieved a measurable 25% average performance improvement among mentored students through targeted, highly personalized academic support plans and comprehensive concept reviews.

AIUB Research & Development Club (R&DC)
Event Manager & Organizer

Dhaka, Bangladesh
Oct 2023 — Dec 2025

- Coordinated, managed, and executed over 15 large-scale technical events, research seminars, and hands-on workshops with a collective attendance exceeding 500 participants.
- Managed multi-department logistics, developed event schedules, and led volunteer teams, successfully enhancing campus-wide undergraduate participation in scholarly research and engineering initiatives.

ESAB AIUB Unit Face
General Member

Dhaka, Bangladesh
Oct 2023 — Dec 2025

- Supported the planning and execution of student activities and events, fostering collaboration and teamwork within the engineering community.
- Contributed to initiatives that enhance student engagement and promote professional development among peers.

ENGINEERING & CAPSTONE RESEARCH PROJECTS

Unmanned Underwater Vehicle (UUV) — Capstone Project
Project Team Leader (Supervised by Prof. Dr. Md. Abdur Rahman, Pro-VC, AIUB)

May 2024 — May 2025

- Led a multi-disciplinary four-member engineering team to design, manufacture, and validate a prototype Remotely Operated Vehicle (ROV) optimized for real-time marine object detection and environmental telemetry.
- Engineered the embedded control and hardware architectures utilizing a Raspberry Pi and a Pixhawk flight controller, integrating multi-sensor arrays and underwater control algorithms.
- Implemented computer vision and real-time sensor networks to enable autonomous object tracking and underwater exploration capabilities.

Vision-Guided Underwater Waste Retrieval — Robotics Project
Project Team Leader (Instructed by Dr. Ebad Zahir, Associate Professor, AIUB)

Feb 2025 — Jun 2025

- Developed and deployed an autonomous underwater vehicle equipped with machine learning algorithms to detect, classify (plastic, metal, organic), and retrieve scattered aquatic debris.
- Designed and integrated a robust robotic grabbing mechanism with vision-guidance feedback loop for automated retrieval, maintaining hardware assembly, simulations and systematic hardware testing.

Synchronous Sequential Multiplier: FSM-Based ALU Design
Digital Design Course Project (Instructed by Dr. Shahriyar Masud Rizvi, Associate Professor, AIUB)

Nov 2024 – Dec 2024

- Designed and simulated a 3×3 unsigned parallel sequential multiplier in SystemVerilog using a Finite State Machine (FSM) and detailed datapath architecture.
- Constructed ASM charts, completed register-transfer level (RTL) coding, and validated operations with an Active-HDL simulation under a 10 MHz verified clock.
- Optimized system performance using a dedicated counter, comparing physical layout area, timing constraints, and scalability metrics.

INDUSTRY EXPERIENCE & TRAINING

Dhaka Electric Supply Company Ltd. (DESCO)
Engineering Intern

Dhaka, Bangladesh
Oct 2025 — Nov 2025

- Analyzed grid operations, power distribution protocols, substation equipment operations, and safety guidelines under professional field engineers.
- Assisted in monitoring power transformer thermal characteristics, electrical load distributions, fault diagnosis, and routine maintenance protocols for substations.

TAHOE (VLSI Training Institute in Bangladesh)
IC Mask Design (Layout) Trainee

Dhaka, Bangladesh
Sep 2025 — Jan 2026

- Successfully completed an industry-standard IC Mask Design (Layout) training program using licensed Cadence EDA tools.
- Covered the complete IC design and layout flow, CMOS layout techniques, post-layout verification, and semiconductor fabrication processes in collaboration with Ulkasemi and the University of Asia Pacific.

Ulkasemi (Semiconductor Design Company)
VLSI Design Trainee

Dhaka, Bangladesh
Oct 2025 — Oct 2025

- Selected among the top 100 engineering students for industrial VLSI training.
- Acquired practical competencies in full custom IC design flow, covering analog and digital circuit design, verification, layout rules, and standard placement and routing protocols.

CORE TECHNICAL COMPETENCIES

Power & Renewable Energy	MATLAB/Simulink (Power, Control & Communication Systems), PVsyst, HOMER Energy, RETScreen, Proteus (PCB), EasyEDA, NI Multisim, CST Microwave Studio, COMSOL Multiphysics
VLSI & Digital Design	Cadence Virtuoso, Analog/Digital Design & Verification, Xilinx Vivado (AMD), Aldec Active-HDL, FPGA Design, FSM Architecture, Verilog, SystemVerilog, VHDL
Embedded Systems & Robotics	Raspberry Pi, Pixhawk Flight Controller, STM32, ESP32, Arduino, Real-time Sensor Integration, Autonomous Control Algorithms
Languages & Academic Tools	Python, C/C++, LaTeX, Technical Writing, Statistical Analysis, MS Office

HONORS & AWARDS

- **Summa Cum Laude with Gold Medal (Academic Award)** : Recognized for outstanding academic achievement maintaining a near-perfect CGPA of 3.97/4.00 throughout the entire undergraduate program.
- **AIUB Dean's List Award (4 Times)** : Awarded to outstanding students for maintaining a CGPA of 3.95-4.00 in each academic year.
- **3rd Place, World Nanotechnology Day Poster Competition (2023)** : Organized jointly by Center for Nanotechnology Research (CNR) and AIUB Community of Engineering Students (ACES).

LANGUAGES

Bengali (Native) | English (Fluent – Professional proficiency in written and spoken communication)

REFERENCES

Prof. Dr. Md. Abdur Rahman
Pro-Vice Chancellor & Professor
Faculty of Engineering, AIUB
(Undergraduate Capstone Project Supervisor)
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Dr. Md. Rifat Hazari
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